

Question Number	Answer	Additional guidance	Mark
8(a)	A description that makes reference to three of the following points: • translation occurs at rER ribosomes / {polypeptide (chain) / chain of amino acids} enters rER (1) • formation of {α-helix / β-pleated sheet / secondary structure / tertiary structure / 3D shape / globular structure} (1) • (causing) formation of {hydrogen bonds / covalent bonds / ionic bonds / disulphide bridges} • {polypeptide / protein / enzyme} packaged into (transport) vesicle to go to Golgi apparatus (1)	ACCEPT rER ribosomes join amino acids together with peptide bonds / forms primary structure / forms polypeptide REJECT if linked to wrong structure ACCEPT vesicle containing {polypeptide / protein / enzyme} {transported to / fuses with} Golgi {apparatus / body}	(3)

Question Number	Answer	Additional guidance	Mark
8(b)(i)	An answer that makes reference to the following points: - as time increases the mean percentage loss in mass increases (1) - there is a higher percentage mass loss with species A (at each time point) / species A digests more wood / {rate of mass loss / digestion} is faster for species A (1) - significant difference between the two species as the {error / range / SD} bars do not overlap / species B is significantly lower than A as the {error / range / SD} bars do not overlap (or converse) (1)	ACCEPT positive correlation ACCEPT as time increases the wood blocks mass decreases ACCEPT species A is more effective at breaking down wood molecules ACCEPT converse for species B ACCEPT correct calculated difference ACCEPT {steeper / non-linear} increase in mass loss in species A ACCEPT converse for species B / rate decreases over time for B ACCEPT correct statements regarding specific species and specific time period {error / range / SD} bars overlap linked to significant difference ACCEPT correct statements regarding specific species and specific time period {error / range / SD} bars size linked to {repeatability / validity} IGNORE bars do not overlap so no significant difference unqualified IGNORE incorrect statements regarding {error / range / SD} e.g. there is overlap of bars between species A& B	(3)

Question Number	Answer	Additional guidance	Mark
8(b)(ii)	An answer that makes reference to the following point: because the species produce different enzymes / species A broke down cellulose as well as lignin (1)	ACCEPT ideas relating to {species A had optimum conditions / species A&B have different optimum temperatures (due to natural habitat)} ACCEPT species A can breakdown the predominant molecule in the wood / converse for species B ACCEPT {more / higher concentration of / more effective} enzymes produced by species A / converse for B ACCEPT described differences in blocks IGNORE amount of enzymes released was different unqualified IGNORE stronger enzymes	(1)

Question Number	Answer	Additional guidance				Mark
8(c)(i)	An answer that includes the following points: - N(N-1) correctly calculated (1) $\sum n(n-1)$ correctly calculated (1) - calculation of D for habitat two (1)		Species	Number of individuals (n)	n(n-1)	(3)
			A	34	1122	
			B	6	30	
			C	27	702	
			D	4	12	
			E	9	72	
			F	31	930	
			G	120	14280	
				N=231	$\sum n(n-1)=17148$	
					D= 3.098	
ALLOW ecf for mp3 if rounded correctly IGNORE fractions correct answer on answer line scores 3 marks						

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8(c)(ii)	• habitat 1 because it has {a higher index of diversity / more species} (1)	E.g. habitat 1 as 4.2 is larger than {3.098 / 3.1 / 3} IGNORE higher biodiversity unqualified ecf for correct conclusion using incorrect D value	(1)

Question Number	Answer	Additional guidance	Mark
8(c)(iii)	An explanation that makes reference to three of the following points: • collection of seeds from multiple plants of species D (to ensure different alleles) (1) • {washing / disinfecting / sterilising} seeds to remove (decomposing) microbes (1) • x-ray seeds to check {viability / presence of embryo} (1) • {freeze the seeds / dry the seeds / store in very low temperatures} to {prevent germination / maintain viability / prevent growth of microbes / reduce enzyme activity / keep them dormant} (1) • {germination / growth / pollination} of genetically different plants to collect new seeds (1)	IGNORE large number of seeds IGNORE collect seeds unqualified ACCEPT collect seeds from different areas reject from different species ACCEPT use antimicrobials to remove microbes IGNORE cool / cold IGNORE {drying / freezing} plants IGNORE {freeze / dry} seeds without explanation ACCEPT {growth / pollination} of plants to collect new seeds ACCEPT {plant/ germinate} some seeds to check viability ACCEPT cloning	(3)